

RAJESH BOTLA

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Guntur, Andhra Pradesh, India

Professional Summary

Computer Science graduate and aspiring Software Engineer with hands-on experience in Java (Spring Boot), Python, Distributed Systems, and AI/ML development. Demonstrated skill in building event-driven microservice architectures, high-performance video streaming platforms, and RAG-based AI applications. Skilled in REST API design, caching, containerization, and data structures.

Technical Skills

Languages: Java 17, Python, JavaScript, HTML5, CSS3, SQL

Frameworks & Libraries: Spring Boot, LangChain, Streamlit, Hugging Face Transformers, HLS.js, PyTest, JUnit

Data & Cloud Infrastructure: MySQL, Redis (Caching), AWS S3, Apache Kafka, Vector Databases (FAISS), NoSQL

Architectural Patterns: Microservices, Event-Driven Architecture, Retrieval-Augmented Generation (RAG), RESTful API Design

Tools & DevOps: Docker, Docker Compose, Git, GitHub, FFmpeg, Tesseract OCR, Postman

Core CS Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design, SOLID Principles

Work Experience

AI & ML Engineer Intern

Dec 2025 – Apr 2026

Blackbucks Education Pvt. Ltd.

Guntur, India

- Engineered end-to-end Python ML pipelines using NumPy and Pandas across 5+ datasets, reducing manual data-preprocessing overhead by ~35%.
- Built a multi-object detection computer vision model with OpenCV, achieving 88% localization accuracy across 10 object classes in real-time video streams.
- Prototyped and optimized CV workflows in Google Colab, leveraging automated image feature extraction to accelerate testing cycles by 40%.
- Completed a structured 16-week internship focused on Python programming, core machine learning workflows, and software development lifecycle (SDLC) best practices.

Projects

Netflix-Inspired Distributed Video Streaming Platform | *Java 17, Spring Boot, Kafka, Redis, AWS S3, Docker*

GitHub Repository

- Architected a Netflix-inspired streaming platform as four independent Spring Boot microservices (Content, Video, Encoding, Streaming), enabling independently deployable, loosely coupled services.
- Engineered an event-driven video processing pipeline with Apache Kafka to orchestrate video upload, HLS encoding, and streaming across services, decoupling processing stages.
- Automated video transcoding with FFmpeg to generate adaptive HLS streams in 1080p, 720p, 480p, and 360p for smooth adaptive-bitrate playback.
- Integrated AWS S3 for scalable video storage, MySQL for persistent data management, and Redis to cache frequently accessed movie metadata and reduce repeated database reads by over 50%.
- Built a responsive frontend in HTML, CSS, and vanilla JavaScript with HLS.js, consuming REST APIs for movie browsing, search, and adaptive streaming.
- Containerized all four microservices along with MySQL, Redis, Kafka, and Zookeeper via Docker Compose, streamlining local development and deployment to a single command.

SparkAI: LLM-Powered Multimodal AI Assistant with RAG | *Python, Streamlit, LangChain, Gemini, FAISS, Docker*

- Built a conversational assistant with a Streamlit interface, persistent local session management, and transcript exporting.
- Integrated Google Gemini and Groq APIs to power fast text generation alongside a Retrieval-Augmented Generation (RAG) pipeline for document-aware responses.
- Built a document ingestion pipeline for PDF and text files covering chunking, embedding generation, vector indexing, and OCR-based extraction (Tesseract) for scanned documents.
- Implemented semantic search with FAISS and Hugging Face sentence transformers, using incremental indexing and document hashing to avoid redundant re-embedding.
- Containerized the application with Docker and configured secure, environment-based API key management.

Achievements

- Awarded **Best Performer** in a 36-hour IoT Hackathon for building an automated environmental monitoring prototype using microcontrollers and sensors to deliver real-time telemetry and threshold alerts.

Education

KITS Akshar Institute of Technology

Oct 2022 – June 2026

B.Tech in Computer Science and Engineering | CGPA: 7.8 / 10.0

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NRI Junior College

March 2022

Intermediate (M.P.C) | Percentage: 75%

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Z.P. High School

March 2020

Secondary Education (SSC) | Percentage: 93%

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Certifications

Introduction to Large Language Models – Simplilearn

July 2026

Artificial Intelligence – Infosys Springboard

January 2025

Java Programming – Infosys Springboard

June 2024

Python Bootcamp – Dev Nirwal

April 2023